Viewing files

head

- view the start of a file
- first 10 lines by default

options:

`-c num` `--bytes=num`	view first **num** bytes
`-n num` `--lines=num`	view first **num** lines

tail

- view the end of a file
- has options same as head and some extra ones:

`-f` `--follow`	keep the file open and display new lines as they're added good for tracking log files

example
combining head with tail
extract the first 15 lines, then display its last 5 lines
`head --n 15 sample.txt | tail --n 5`

Transforming files

These commands don't change files contents, but instead send the changed file contents to STDOUT

expand

converting tabs to spaces

unexpand

convert spaces to tabs

sort

sort a file

```
james@laptop:~/tmp$ cat capital
Itanagar
Dispur
Hyderabad
Patna
Raipur
james@laptop:~/tmp$ sort capital
Dispur
Hyderabad
Itanagar
Patna
Raipur
```

options

Ignore case	`-f` `--ignore-case` don't differentiate uppercase/lowercase
Month	`-M` `--month-sort` sort by three letter month abbreviation e.g. JAN
Numeric	`-n` `--numeric-sort` sort by number
Reverse	`-r` `--reverse-order` sorts in reverse order
Field	`-k` `--k=field` by default uses the first field as its sort field, you can specify another field/s

```
$ sort -k 3 listing1.1.txt
555-2397 Beckett, Barry
555-5116 Carter, Gertrude
555-9871 Orwell, Samuel
555-7929 Jones, Theresa
```

split

split a file into multiple files

options

Split by bytes	`-b size` `--bytes=size` breaks input file into pieces by maximum *size* bytes
by bytes in line-size chunks	`-C=size` `--line-bytes=size` same as split by bytes, but without breaking lines
by number of lines	`-l lines` `--lines=lines`

example - split listing1 into two files – *numbersaa* and *numbersab*

```
james@laptop:~/tmp$ cat listing1
555-2397 Beckett, Barry
555-5116 Carter, Gertrude
james@laptop:~/tmp$ split -l 1 listing1 numbers
james@laptop:~/tmp$ ls
capital  listing1  listing2  numbersaa  numbersab  state
```

tr

translate (replace)

- replace characters in a SET1 with characters in SET2
- each character in SET1 is replaced by the one at the equivalent position in SET2

`tr [options] set1 [set2]` translate characters from SET1 to SET2

options

`-t` `--truncate-set1`	truncate SET1 to size of SET2
`-d`	delete characters from SET1, no need for a SET2

example: B→b, C→c, G→c

note that set2 was smaller than set 1, it substitutes the last letter from set2 for the missing letter

```
james@laptop:~/tmp$ cat listing1
555-2397 Beckett, Barry
555-5116 Carter, Gertrude
james@laptop:~/tmp$ tr BCG gc < listing1
555-2397 geckett, garry
555-5116 carter, certrude
```

shortcuts

`:alnum`	truncate SET1 to size of SET2
`:upper` `:lower`	delete characters from SET1, no need for a SET2
`:digits`	delete characters from SET1, no need for a SET2

examples

`echo "hello world 123" | tr '[:lower:]' '[:upper:]'` converts all lowercase to upper

`echo "password" | tr '[:alnum:]' '*'` replace all characters with *

`echo "hello, world! 123" | tr -d [:punc:][:space:]` delete punctuation and whitespace

uniq

delete duplicate lines

e.g. if you've sorted a file and don't want duplicates e.g. Shakespeares vocabulary

```
james@laptop:~/tmp$ cat file1
apple
apple
pear
james@laptop:~/tmp$ cat file2
james@laptop:~/tmp$ sort file1 | uniq > file2
james@laptop:~/tmp$ cat file2
apple
pear
```

od

octal dump

- For displaying files that are not easily displaYed in ASCII
- e.g. audio files etc will look like gibberish
- displays a file in octal (base 8) numbers by default

```
james@laptop:~/tmp$ od listing1
0000000 032465 026465 031462 033471 041040 061545 062553 072164
0000020 020054 060502 071162 005171 032465 026465 030465 033061
0000040 041440 071141 062564 026162 043440 071145 071164 062165
0000060 005145
0000062
```

Combining Files

cat

concatenate

display:

`cat first.txt Data from first file`

to file:

`cat first.txt second.txt > combined.txt`

options

Display line ends	`-E` `--show-ends` add **$** (dollar sign) at the end of each line
Number lines	`-n` `-number` add line number to start each line
Minimize blank lines	`-s` `--squeeze-blank` Compresses groups of blank lines to a single line

tac

Same as cat but it reverses the order of lines in the output

```
$ cat combined.txt
Data from first file.
Data from second file.
$
$ tac combined.txt
Data from second file.
Data from first file.
```

join

Like a database join

By default, uses the first field as they key to match across files

```
james@laptop:~/tmp$ cat listing1
555-2397 Beckett, Barry
555-5116 Carter, Gertrude
james@laptop:~/tmp$ cat listing2
555-2397 unlisted
555-5116 listed
james@laptop:~/tmp$ join listing1 listing2
555-2397 Beckett, Barry unlisted
555-5116 Carter, Gertrude listed
```

Can use another field as the key

`join -1 3 -2 2 cameras.txt lenses.txt`

join using **3rd** field in cameras and **2nd** field in lenses

paste

Merge files line by line, separating each line with a tab

```
$ paste listing1.1.txt listing1.2.txt
555-2397 Beckett, Barry       555-2397 unlisted
555-5116 Carter, Gertrude     555-5116 listed
555-7929 Jones, Theresa       555-7929 listed
555-9871 Orwell, Samuel       555-9871 unlisted
```

Formatting files

Commands that reformat the text in a file

fmt

- limit text files width
- 75 characters by default

options:

`-w width` `--width=width`	set line length to width characters

nl

numbers non-blank lines of a file
options:

numbering style
`-b style-option`
`--body-number=style-option`
	style-option
	t - default, numbers lines except empty
	a - number all lines including empty one

new page `-p` `--no-renumber`	default is to number each new page starting with 1, the options don't reset the line number with a new page
page separator `-d=code` `--section-delimiter-code`	**code** is the delimeter that defines a new page

example
you get error messages that refer to a script line
to add line numbers to the script

```
nl –b a somescript > numberedscript.txt
```

pr

prepare a file for printing
by default creates a header which includes:

- current date/time
- filename
- page number

options:

multi-column output
`-numcols`
`--columns=numcols`
	creates output with **numcols** columns

double spacing
`-d`
`--double-spaced`
	double spaced output

page length
`-l lines`
`--length-lines`
	page length in **lines**

header text
`-h text`
`--header=text`
	set text to be displayed in header, replacing the filename

omit header `-t` `--omit-header`	

left margin
`-o chars`
`--indent=chars`
	sets left margin to **chars** characters

page width
`-w chars`
`--width cars`	

example

`cat –n /etc/profile | pr –d | lpr`

cat –n output with line numbers
pr –d double space
lpr print the file

Summarizing files

cut

cut/extract certain parts of each line and display them
options:

`-b list` `--bytes=list`	cut first **list** bytes
`-c list` `--characters=list`	cut first **list** characters
`-f list` `--fields=list`	cut the specified list of fields

list option

- a number e.g. 4
- a range of numbers e.g. 2-4

wc

word count
counts words, lines and bytes for a file

```
james@laptop:~/tmp$ cat listing1
555-2397 Beckett, Barry
555-5116 Carter, Gertrude
james@laptop:~/tmp$ wc listing1
 2  6 50 listing1
```